Micro Hack 2 — SDC Workloads

Participant Workbook · GreenGrid Scorecard

FY27 EMEA EPS · Fabric Apps Motion

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Micro Hack 2 — SDC Workloads Workbook (Participants)

Track: **SDC Workloads (Fabric Extensibility Toolkit)** Scenario: **GreenGrid Analytics** (fictional SDC)

Print-ready PDF: `microhack-2-sdc-workloads-workbook.pdf`

This is a **didactic** micro-hack. You are **not** expected to *find* the solution — every step tells you **exactly what to type or paste**, **why**, and **what you should see**. Follow them in order and you get a working Fabric workload, starting from a clean machine.

Day agenda (1 day)

Morning — presentations & demos. Afternoon — hands-on hack. This is a **standalone one-day micro-hack for the SDC Workloads (Extensibility Toolkit) track**.

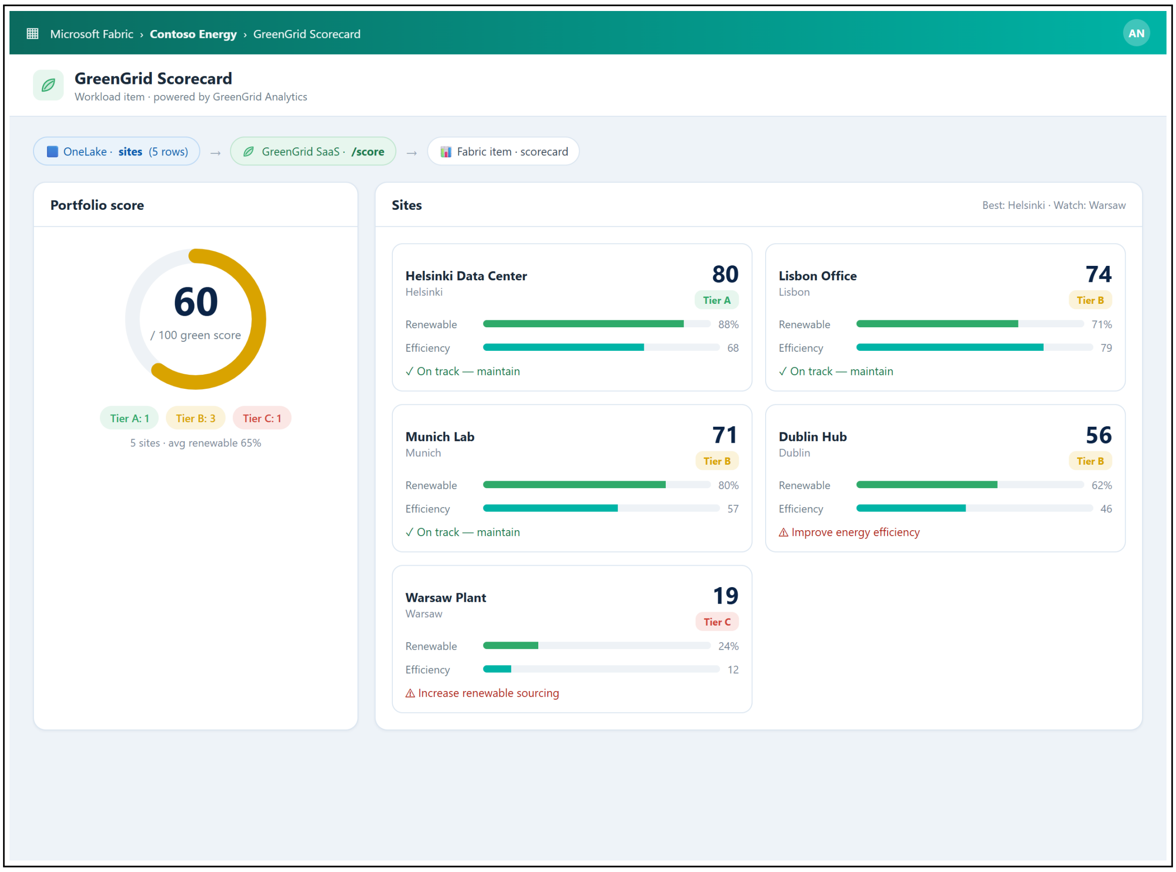
Time	Session
09:00	Welcome & motion overview — the Fabric apps motion for EMEA
09:30	Microsoft Fabric foundations — OneLake, capacity, workspaces
10:00	Extensibility Toolkit foundations — workloads, items, Dev Gateway
10:30	Break
10:45	**Demo · GreenGrid Scorecard** — the workload you'll build + the SaaS prerequisite
11:15	The hack brief, teams & environment check
12:00	Lunch
13:30	**Hack · Sprint 1** — clone, run Hello World, render with fake data (steps 1–7)
15:30	Break
15:45	**Hack · Sprint 2** — switch to OneLake & polish (steps 8–9)
16:45	Team demos (5 min each)
17:00	Wrap-up, KPIs & next steps

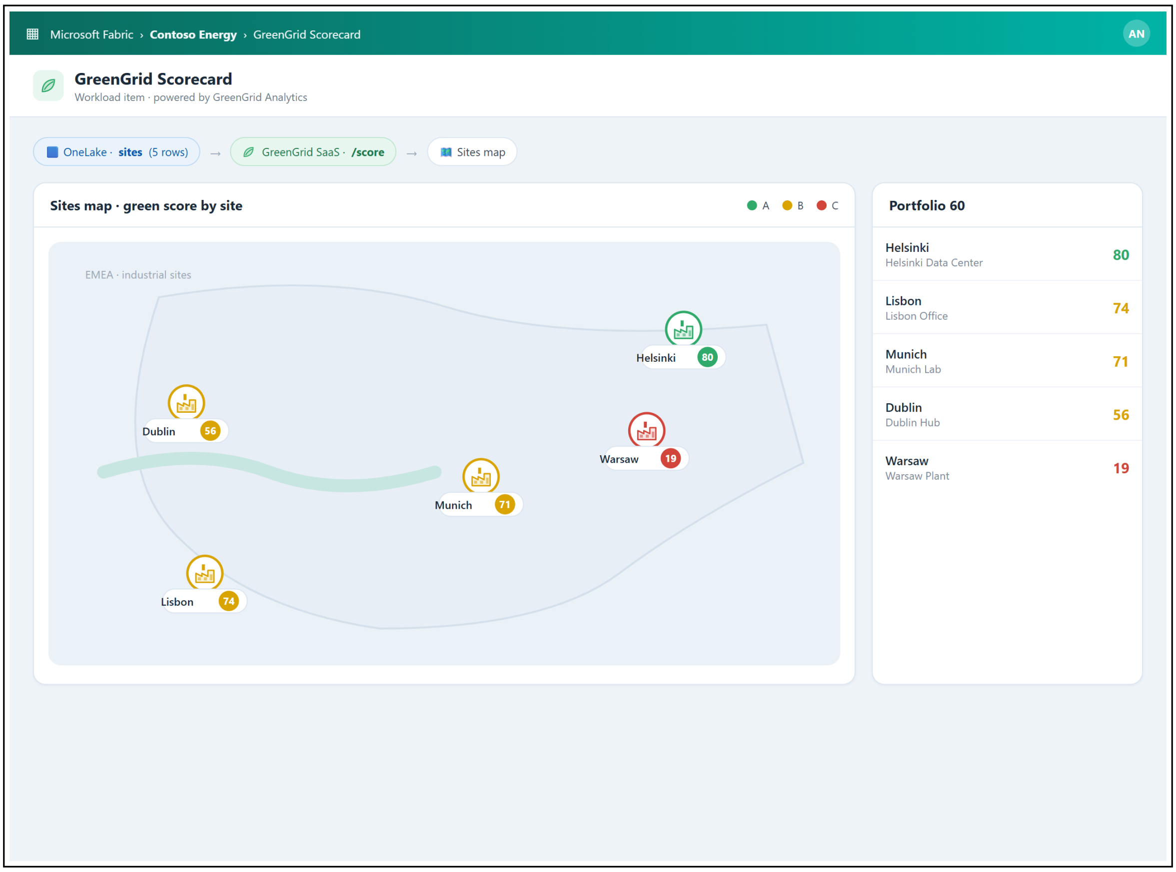
Part A — Understand

1) The scenario in plain words

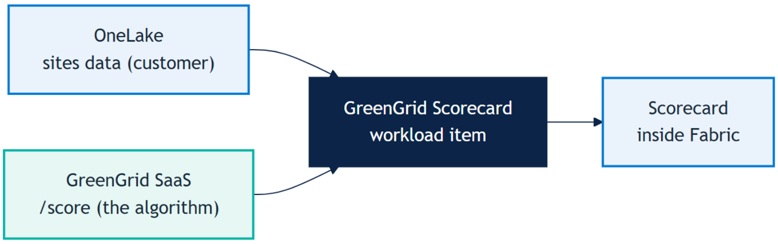
GreenGrid Analytics is a software company (an **SDC** — a solution development company). Its product scores how **sustainable** a customer's sites are, from their energy use and renewable mix. The customer's data already lives in Microsoft Fabric (**OneLake**). Your job: bring GreenGrid's scoring **inside Fabric**, as a native screen, **without moving the data**.

The result looks like this — a sustainability scorecard and a map of the industrial sites, rendered inside Fabric:





2) The three pieces (and who owns the algorithm)



- **OneLake** = where the **customer's data** lives.
- **GreenGrid SaaS** = a small web service that **holds the scoring algorithm** (GreenGrid's IP). *The trainer deploys it for you — it is a prerequisite.*
- **The workload** = the piece **you build today**: a Fabric *item* that reads OneLake, calls the SaaS, and draws the scorecard.

3) Key words (30-second primer)

- **Workload**: a web app that runs *inside* the Fabric portal. You host it, Fabric shows it.
- **Item**: one thing a user creates in a workspace (here: a *GreenGrid Scorecard*).
- **Extensibility Toolkit**: Microsoft's open-source kit + SDK to build such workloads.
- **Dev Gateway**: a local bridge so Fabric can show your workload while it runs on **your machine** — this is what makes a one-day hack possible.
- **OBO token** (On-Behalf-Of): a secure token the workload gets so it can read the user's OneLake data **as that user** — no passwords, no data copies.

4) Prerequisites & environment setup (do this first)

Modeled on Microsoft's Build LAB514 setup. On a lab VM most tools are pre-installed.

Tools you need:

- Node.js 20+ and npm
- **PowerShell 7**, **.NET (x64)**, **Azure CLI** (the toolkit setup script uses them)
- Visual Studio Code
- **GitHub Copilot CLI** (the `copilot` terminal agent) — used for the graphical polish step

Sign in (two accounts):

1. **GitHub Copilot** — run `copilot`, then `/login`, choose **GitHub.com**, finish the browser sign-in, confirm *"Signed in successfully"*, then `/exit`.
2. **Microsoft Fabric** — open `https://app.fabric.microsoft.com` and sign in. *(If it shows "Power BI" bottom-left, switch it to "Fabric".)*

From the trainer / your tenant:

- A **Fabric workspace** with a **capacity** assigned (where the workload runs).
- Permission to create an **Entra app** (or one provided) — required even with the Dev Gateway; `Setup.ps1` creates it.
- The **GreenGrid SaaS URL + API key**.

Verify your setup:

```
node --version
npm --version
copilot --version
pwsh --version
```

What you should see. All print a version, you're signed in to GitHub Copilot and Fabric, and you have the SaaS URL + key in hand.

5) Team framing (fill this in 2 minutes, then build)

- Team name:
- SaaS URL + API key:
- One sentence you'll say in the demo:

Part B — Build it step by step (full solution)

> **Golden rule (read this!):** first get the toolkit running with the built-in **Hello World** (steps 1–4). Only then build GreenGrid (steps 5–9). And inside GreenGrid, render **fake data first** (step 7), *then* switch to real OneLake data (step 8). Never debug security before something shows on screen.

Every step says **what it is**, **what to type/paste**, and **what you should see**.

Step 1 — Clone the toolkit

What this is. Get Microsoft's Extensibility Toolkit on your machine. This is a real repository — you clone it and start from it.

```
git clone https://github.com/microsoft/fabric-extensibility-toolkit
cd fabric-extensibility-toolkit
```

What you should see. A `fabric-extensibility-toolkit` folder with `Workload/` and `scripts/` inside.

Step 2 — Configure the dev environment (one script)

What this is. A setup script that creates the Entra app, writes your `.env`, and downloads the Dev Gateway. It asks you to sign in and pick your workspace.

```
cd scripts/Setup
./Setup.ps1 -WorkloadName "Org.GreenGrid"
```

> The `WorkloadName` must look like `Organization.Name` — use `Org.GreenGrid` for the hack. On macOS/Linux run `pwsh ./Setup.ps1 -WorkloadName "Org.GreenGrid"`.
>
> **Why an Entra app even with the Dev Gateway?** The Dev Gateway only **routes** your local workload into Fabric — it does **not** provide identity. Your workload authenticates with **Microsoft Entra**, and the Entra app is what lets it receive Fabric tokens and acquire the **OBO token** to read OneLake (step 8). `Setup.ps1` creates this app for you (or reuses an existing one).

What you should see. The script ends with "setup complete" and prints the next steps.

Step 3 — Start the dev server + the Dev Gateway

What this is. Two long-running processes, in **two terminals**. One serves your workload UI from localhost; the other connects Fabric to that localhost.

```
# Terminal 1 — the frontend + APIs
cd scripts/Run
./StartDevServer.ps1
```

```
# Terminal 2 — the bridge to Fabric
cd scripts/Run
./StartDevGateway.ps1
```

Then, **in the Fabric portal**: open **Settings -> Admin portal** and enable the developer tenant settings, then **Settings -> Developer settings -> enable Fabric Developer Mode**.

What you should see. Both terminals keep running without errors, and Developer Mode is ON in Fabric.

> **If `StartDevGateway.ps1` fails with** `Dev instance registration ... Forbidden, errorCode: FeatureNotAvailable`: your tenant hasn't enabled customer-developed workloads. An **admin** must turn on, under **Admin portal -> Tenant settings -> Additional workloads**: *Capacity admins and contributors can add and remove additional workloads*, *Workspace admins can develop workloads*, and *Users can see and work with additional workloads not validated by Microsoft* — **and** the workspace must be on a **Fabric/Trial capacity** (not Pro/PPU). Full steps in the setup guide (§6.1). It is not a code issue — the gateway is refused by Fabric.
>
> **Not a tenant admin?** You can't self-enable it — either ask your admin for those toggles, run the hack in a tenant **you** administer (a Fabric Trial / demo tenant), or keep building locally without Fabric: `npm run saas:start` + `npm run demo:build`, then open `src/workloadsdc/demo/out/scorecard.html` (same algorithm, no Fabric needed). See setup §6.1.1.

Step 4 — Hello World test (prove the gateway works, before building anything)

What this is. The toolkit ships a **Hello World** item. Create it in Fabric to confirm the whole chain — *your localhost code -> Dev Gateway -> Fabric* — works. **This is your safety checkpoint.**

1. Go to `https://app.fabric.microsoft.com/workloadhub/detail/Org.GreenGrid.Product?experience=developer` (the workload shows as "Hello Fabric!" until you rename it).
2. Click the **Hello World** item type on the left.
3. Pick your **dev workspace** in the dialog.
4. The Hello World editor opens.

What you should see. The Hello World editor renders **inside Fabric**, served from your machine. If you see it, everything works — now you can build GreenGrid. *(Nothing shows up? Developer Mode is off, or one of the two terminals isn't running.)*

Step 5 — Create the GreenGrid item

What this is. Generate a fresh item to hold your scorecard (don't overwrite Hello World).

```
cd scripts/Setup
./CreateNewItem.ps1 -Name "GreenGridScorecard"
```

This creates `Workload/app/items/GreenGridScorecardItem/`, including `GreenGridScorecardItemEditor.tsx`. **Restart the dev server** (Ctrl+C in Terminal 1, then `./StartDevServer.ps1` again). In Fabric, create a **GreenGridScorecard** item.

What you should see. An empty GreenGridScorecard editor opens in Fabric.

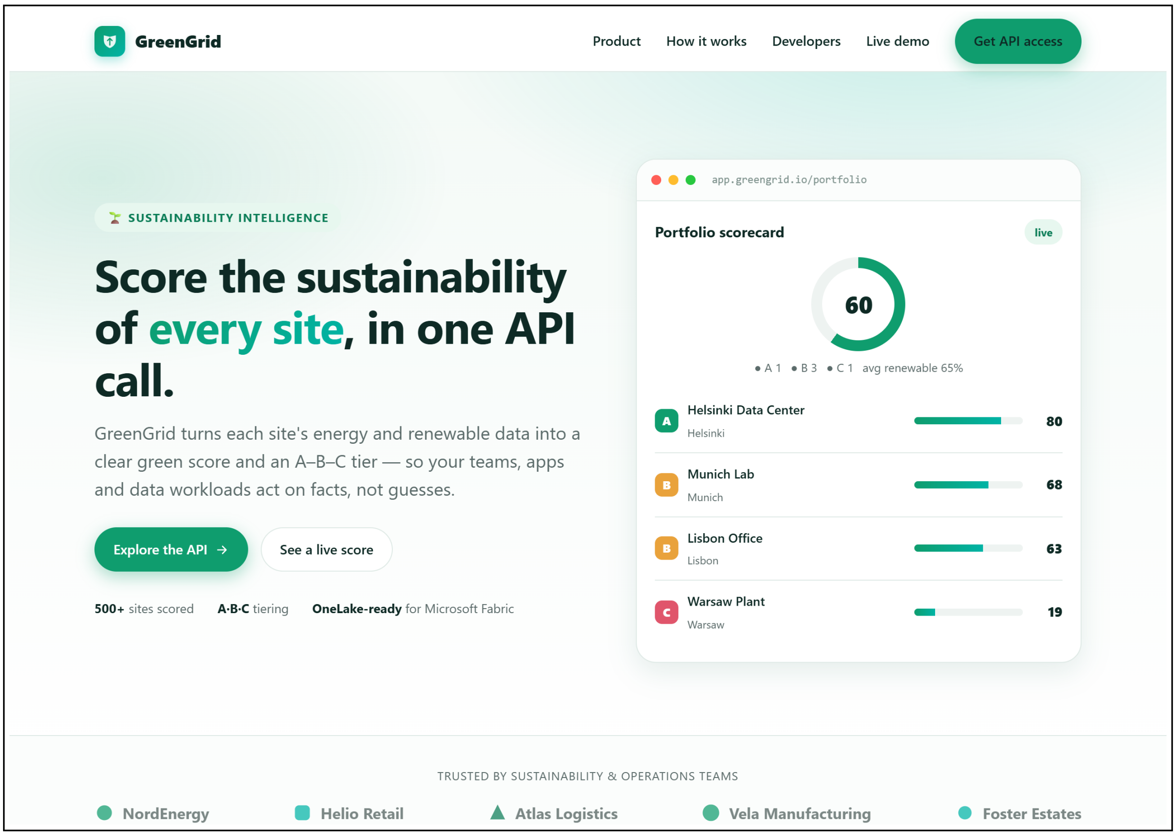
Step 6 — Check the SaaS is alive

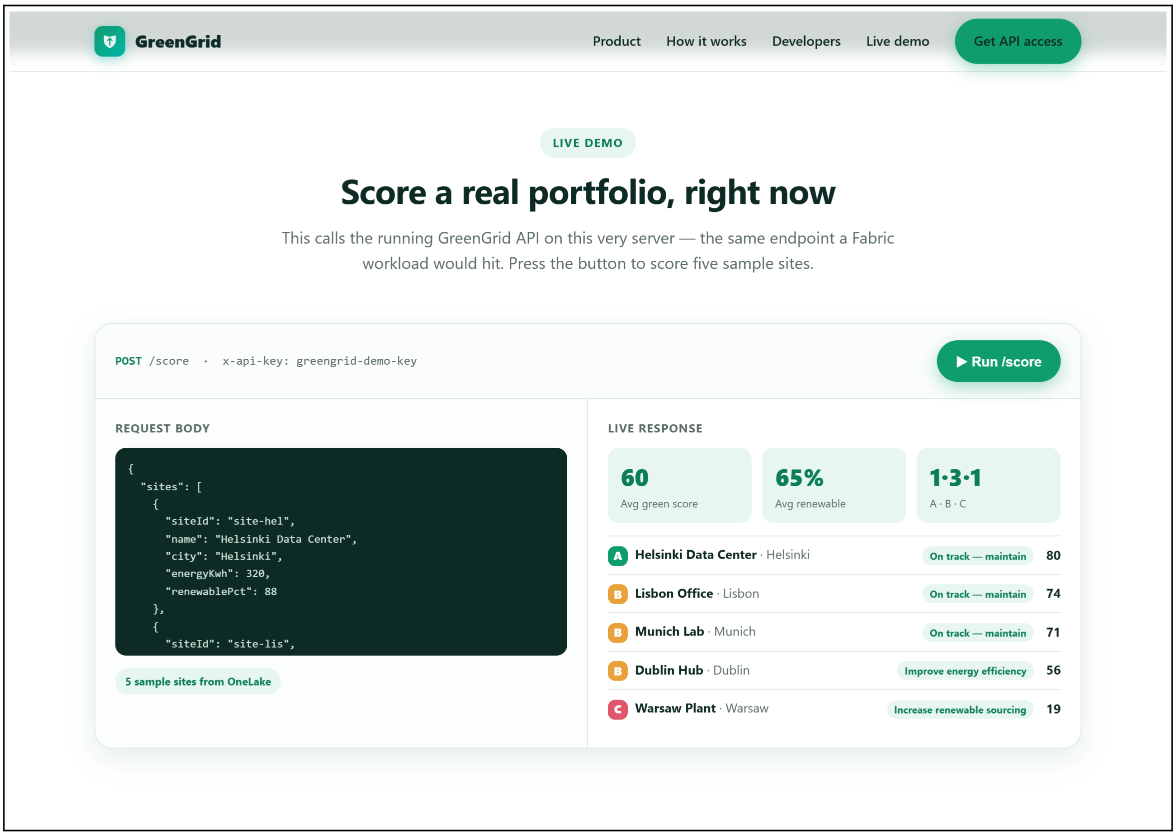
What this is. A quick test that GreenGrid's algorithm service answers.

```
curl -i <SAAS_URL>/health        # expect: 200
curl -s -X POST <SAAS_URL>/score \
  -H "Content-Type: application/json" -H "x-api-key: <API_KEY>" \
  -d '{"sites":[{"siteId":"s1","name":"Helsinki DC","city":"Helsinki","energyKwh":320,"renewablePct":88}]}'
```

What you should see. `/score` returns a `greenScore` and a `tier`. *(Local fallback: `cd src/workloadsdc && npm install && npm run saas:start`, URL `http://localhost:8787`, key `greengrid-demo-key`.)*

Open `<SAAS_URL>/` (e.g. `http://localhost:8787/`) in a browser: the SaaS also serves the **GreenGrid website** with a *Developers* section and a **live demo** that calls `/score`. It's the same scoring algorithm your workload is about to consume.





Step 7 — Make it work with fake data (Milestone M1)

Create **four files** in `Workload/app/items/GreenGridScorecardItem/`, then render the scorecard from the generated editor. This proves the whole chain *screen -> SaaS -> screen*.

7a. `contracts.ts` — the data shapes

```
export type SiteRecord = {
  siteId: string; name: string; city: string; energyKwh: number; renewablePct:
    number;
};
export type GreenTier = 'A' | 'B' | 'C';
export type ScoredSite = SiteRecord & {
  efficiency: number; greenScore: number; tier: GreenTier; tip: string;
};
export type ScoreResponse = {
  sites: ScoredSite[];
  summary: { totalSites: number; avgScore: number; avgRenewablePct: number;
    tierCounts: Record<GreenTier, number>; best: ScoredSite; worst: ScoredSite; };
};
```

7b. `greengridClient.ts` — call the SaaS

```
import type { ScoreResponse, SiteRecord } from './contracts';

const SAAS_URL = 'http://localhost:8787';    // <- replace with the trainer URL
const API_KEY = 'greengrid-demo-key';        // <- replace with the trainer key

export async function scorePortfolio(sites: SiteRecord[]): Promise<ScoreResponse> {
  const res = await fetch(`${SAAS_URL}/score`, {
    method: 'POST',
```

```
    headers: { 'Content-Type': 'application/json', 'x-api-key': API_KEY },
    body: JSON.stringify({ sites }),
  });
  if (!res.ok) throw new Error(`GreenGrid SaaS error ${res.status}`);
  return res.json();
}
```

7c. `seed.ts` — fake sites to start

```
import type { SiteRecord } from './contracts';
export const seedSites: SiteRecord[] = [
  { siteId: 'site-hel', name: 'Helsinki Data Center', city: 'Helsinki', energyKwh:
↪ 320, renewablePct: 88 },
  { siteId: 'site-lis', name: 'Lisbon Office', city: 'Lisbon', energyKwh: 210,
↪ renewablePct: 71 },
  { siteId: 'site-muc', name: 'Munich Lab', city: 'Munich', energyKwh: 430,
↪ renewablePct: 80 },
  { siteId: 'site-dub', name: 'Dublin Hub', city: 'Dublin', energyKwh: 540,
↪ renewablePct: 62 },
  { siteId: 'site-waw', name: 'Warsaw Plant', city: 'Warsaw', energyKwh: 880,
↪ renewablePct: 24 },
];
```

7d. `Scorecard.tsx` — the screen

It takes a `getSites` function, so swapping the data source later is **one line**.

```
import React, { useEffect, useState } from 'react';
import { FluentProvider, webLightTheme, Spinner, Badge } from
↪ '@fluentui/react-components';
import { scorePortfolio } from './greengridClient';
import type { ScoreResponse, SiteRecord } from './contracts';

const tierColor = (t: 'A' | 'B' | 'C') => (t === 'A' ? '#2faa6a' : t === 'B' ?
↪ '#d9a300' : '#d1453b');

export function Scorecard({ getSites }: { getSites: () => Promise<SiteRecord[]> }) {
  const [data, setData] = useState<ScoreResponse | null>(null);
  const [error, setError] = useState<string | null>(null);

  useEffect(() => {
    getSites().then(scorePortfolio).then(setData).catch((e) =>
↪ setError(String(e)));
  }, [getSites]);

  if (error) return <div role="alert">Failed: {error}</div>;
  if (!data) return <Spinner label="Scoring sites..." />;

  return (
    <FluentProvider theme={webLightTheme}>
      <div style={{ padding: 24 }}>
        <h1>GreenGrid Scorecard</h1>
        <p>Data from OneLake, scored by GreenGrid</p>
        <div style={{ fontSize: 48, fontWeight: 700 }}>{data.summary.avgScore}<span
↪ style={{ fontSize: 18 }}> / 100</span></div>
        <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2,1fr)', gap:
↪ 12, marginTop: 16 }}>
          {data.sites.map((s) => (
            <div key={s.siteId} style={{ border: '1px solid #dbe6f1', borderRadius:
↪ 12, padding: 14 }}>
```

```
              <div style={{ display: 'flex', justifyContent: 'space-between' }}>
                <strong>{s.name}</strong>
                <Badge appearance="filled" style={{ background: tierColor(s.tier)
 ↪  }}>Tier {s.tier}</Badge>
              </div>
              <div style={{ color: '#7a8a9a', fontSize: 12 }}>{s.city}</div>
              <div style={{ fontSize: 26, fontWeight: 700 }}>{s.greenScore}</div>
              <div style={{ fontSize: 13 }}>{s.tip}</div>
            </div>
          ))}
        </div>
      </div>
    </FluentProvider>
  );
}
```

7e. Render it from the generated editor

Open `GreenGridScorecardItemEditor.tsx` and render the `Scorecard` with the **fake** data:

```
import { Scorecard } from './Scorecard';
import { seedSites } from './seed';

// inside the editor's returned JSX:
<Scorecard getSites={() => Promise.resolve(seedSites)} />
```

What you should see — Milestone M1. The item shows the portfolio score and 5 site cards with their tier, **inside Fabric**. The whole chain works on fake data. This is demoable.

Step 8 — Switch to real OneLake data (Milestone M2)

What this is. Replace the fake `seedSites` with the customer's **real** site data in OneLake.

Where does the data live?

In Fabric, customer data lives in a **Lakehouse** inside OneLake. A Lakehouse has **Files** (raw files: CSV, Parquet…) and **Tables** (Delta). For this hack the sites are a **CSV file**: `Files/sites.csv`. *(A CSV is the simplest thing to read from a workload; a Delta table would need a SQL endpoint — out of scope for one day.)*

Format of `Files/sites.csv` (a ready copy is in the kit at `src/workloadsdc/data/sites.csv`):

```
siteId,name,city,energyKwh,renewablePct
site-hel,Helsinki Data Center,Helsinki,320,88
site-lis,Lisbon Office,Lisbon,210,71
site-muc,Munich Lab,Munich,430,80
site-dub,Dublin Hub,Dublin,540,62
site-waw,Warsaw Plant,Warsaw,880,24
```

> The trainer pre-loads this file (Lakehouse -> **Files** -> **Upload**). To test your own, upload a CSV with the **same header** to `Files/`.

You also need two IDs (visible in the Lakehouse URL, or from the item context): `workspaceId` and `lakehouseId`.

8a. `onelake.ts` — read & parse the CSV

```
import type { WorkloadClientAPI } from '@ms-fabric/workload-client';
import type { SiteRecord } from './contracts';

export async function readSites(
```

```
  client: WorkloadClientAPI, workspaceId: string, lakehouseId: string
): Promise<SiteRecord[]> {
  // 1) Token to read OneLake AS the signed-in user (OBO).
  const { token } = await client.auth.acquireAccessToken({
    additionalScopesToConsent: ['https://storage.azure.com/user_impersonation'],
  });
  // 2) Read the CSV file from the Lakehouse "Files" area.
  const url = `https://onelake.dfs.fabric.microsoft.com/${workspaceId}/${
  ↪  lakehouseId}/Files/sites.csv`;
  const res = await fetch(url, { headers: { Authorization: `Bearer ${token}` } });
  if (!res.ok) throw new Error(`OneLake read failed: ${res.status}`);
  // 3) Parse CSV text into rows.
  const [header, ...lines] = (await res.text()).trim().split(/\r?\n/);
  const cols = header.split(',').map((c) => c.trim());
  return lines.filter(Boolean).map((line) => {
    const cells = line.split(',');
   const row = Object.fromEntries(cols.map((c, i) => [c, (cells[i] ?? '').trim()]));
    return { siteId: row.siteId, name: row.name, city: row.city,
      energyKwh: Number(row.energyKwh), renewablePct: Number(row.renewablePct) };
  });
}
```

8b. Swap one line in the editor

The generated editor exposes the workload client and the item's context. Change the render to:

```
import { readSites } from './onelake';

// before (fake):
<Scorecard getSites={() => Promise.resolve(seedSites)} />

// after (real OneLake CSV):
<Scorecard getSites={() => readSites(workloadClient, workspaceId, lakehouseId)} />
```

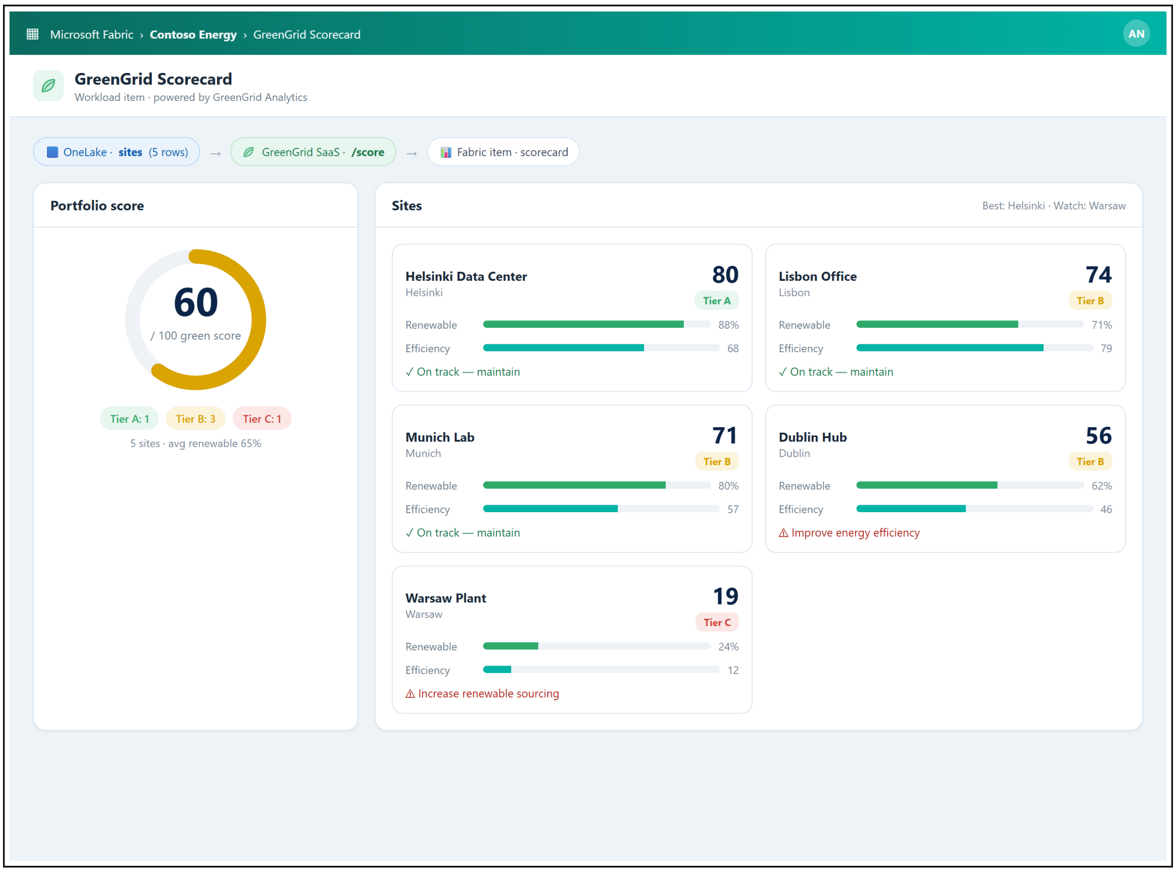
What you should see — Milestone M2. The same scorecard, now built from the **OneLake CSV**. *(If the OBO token fights you, keep the fake-data line as a fallback and keep moving.)*

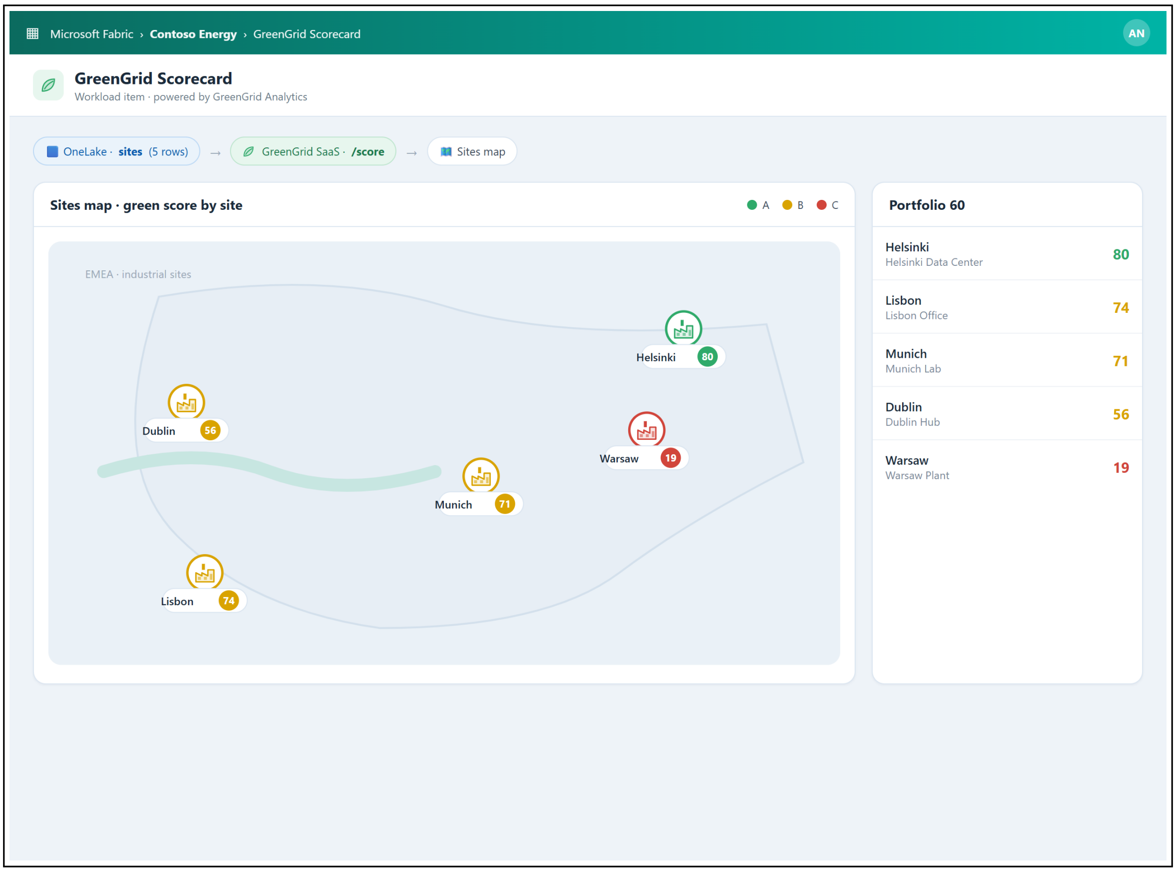
Step 9 — Make it graphical (Milestone M3)

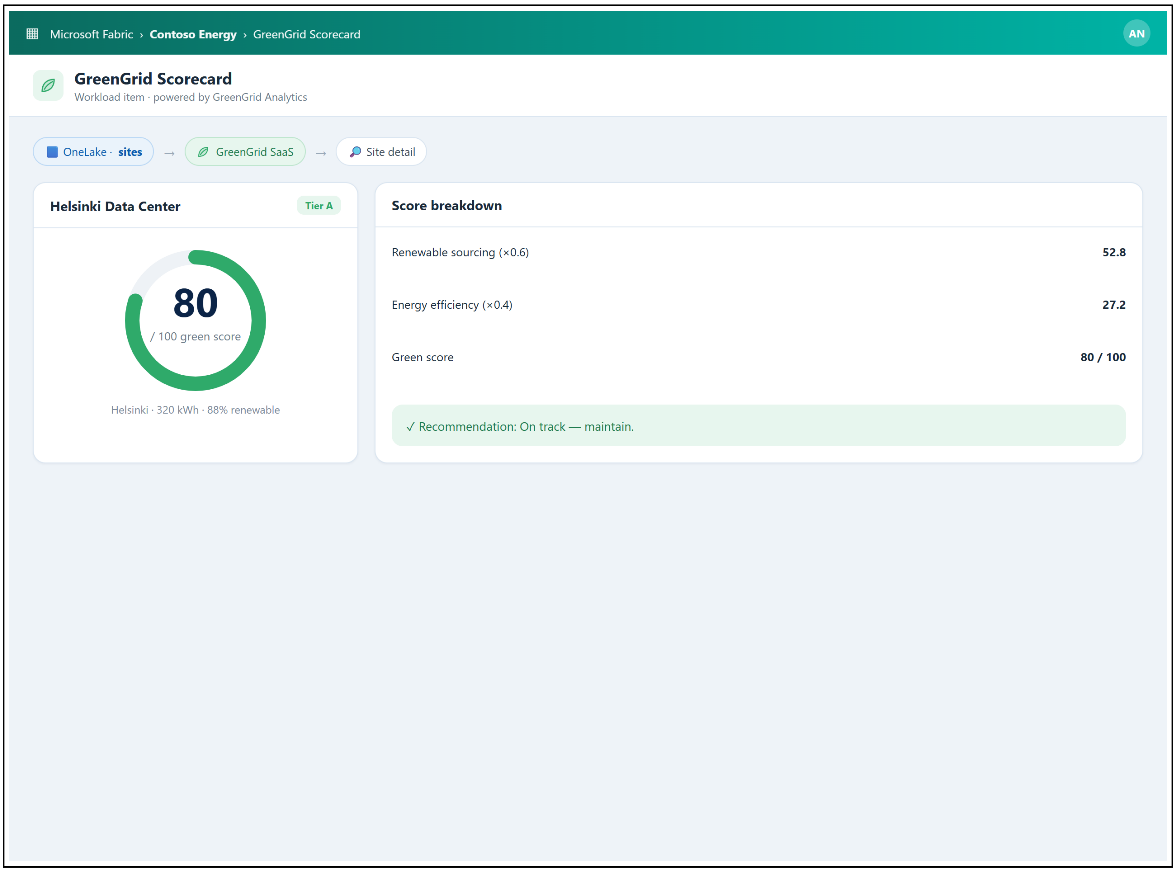
What this is. Upgrade the plain list into the polished scorecard + map. Easiest with GitHub Copilot (the toolkit ships AI instructions in `.github/copilot-instructions.md`):

```
Upgrade the GreenGrid Scorecard screen: add a circular green-score gauge for the
↪  portfolio,
an A/B/C tier distribution, provenance chips "Data from OneLake" then "Scored by
↪  GreenGrid",
and a second view showing the sites as factory markers on a simple map, colored by
↪  tier.
Style: Fluent UI, teal #00b4a6 + green accents. Keep the getSites data flow.
```

What you should see — Milestone M3. A gauge, A/B/C tiers and the sites map, rendered natively inside Fabric:


Microsoft Fabric › Contoso Energy › GreenGrid Scorecard
AN
GreenGrid Scorecard
Workload item · powered by GreenGrid Analytics
OneLake · sites (5 rows) → GreenGrid SaaS · /score → Fabric item · scorecard
Portfolio score
60
/ 100 green score
Tier A: 1 Tier B: 3 Tier C: 1
5 sites · avg renewable 65%
Sites
Best: Helsinki · Watch: Warsaw
Helsinki Data Center 80
Helsinki
Tier A
Renewable 88%
Efficiency 68
✓ On track — maintain
Lisbon Office 74
Lisbon
Tier B
Renewable 71%
Efficiency 79
✓ On track — maintain
Munich Lab 71
Munich
Tier B
Renewable 80%
Efficiency 57
✓ On track — maintain
Dublin Hub 56
Dublin
Tier B
Renewable 62%
Efficiency 46
⚠ Improve energy efficiency
Warsaw Plant 19
Warsaw
Tier C
Renewable 24%
Efficiency 12
⚠ Increase renewable sourcing


Microsoft Fabric › Contoso Energy › GreenGrid Scorecard
AN
GreenGrid Scorecard
Workload item · powered by GreenGrid Analytics
OneLake · sites (5 rows) → GreenGrid SaaS · /score → Sites map
Sites map · green score by site
A B C
EMEA · industrial sites
Helsinki 80
Dublin 56
Warsaw 19
Munich 71
Lisbon 74
Portfolio 60
Helsinki
Helsinki Data Center
80
Lisbon
Lisbon Office
74
Munich
Munich Lab
71
Dublin
Dublin Hub
56
Warsaw
Warsaw Plant
19



Wrap-up

Demo storyboard (5 minutes)

1. **Context** (30s): the SaaS value (the algorithm) you brought into Fabric.
2. **Flow** (3m): open the item, read OneLake, call the SaaS, show the scorecard + map.
3. **Architecture** (45s): OneLake (data) + SaaS (algorithm) + workload (native UI).
4. **Publish plan** (45s): from Dev Gateway to a released workload.

Reflection

- Where did the OneLake -> SaaS contract surprise you?
- What stays in the SaaS (the IP) vs. in the workload (the experience)?
- What would you harden before production?

Reference assets

- Full solution code: `src/workloadsdc/`
- SaaS prerequisite: `src/workloadsdc/saas/`
- Sample OneLake data: `src/workloadsdc/data/sites.csv`
- Trainer answer key + screenshots: `docs/microhack-2-sdc-workloads-solution.md`